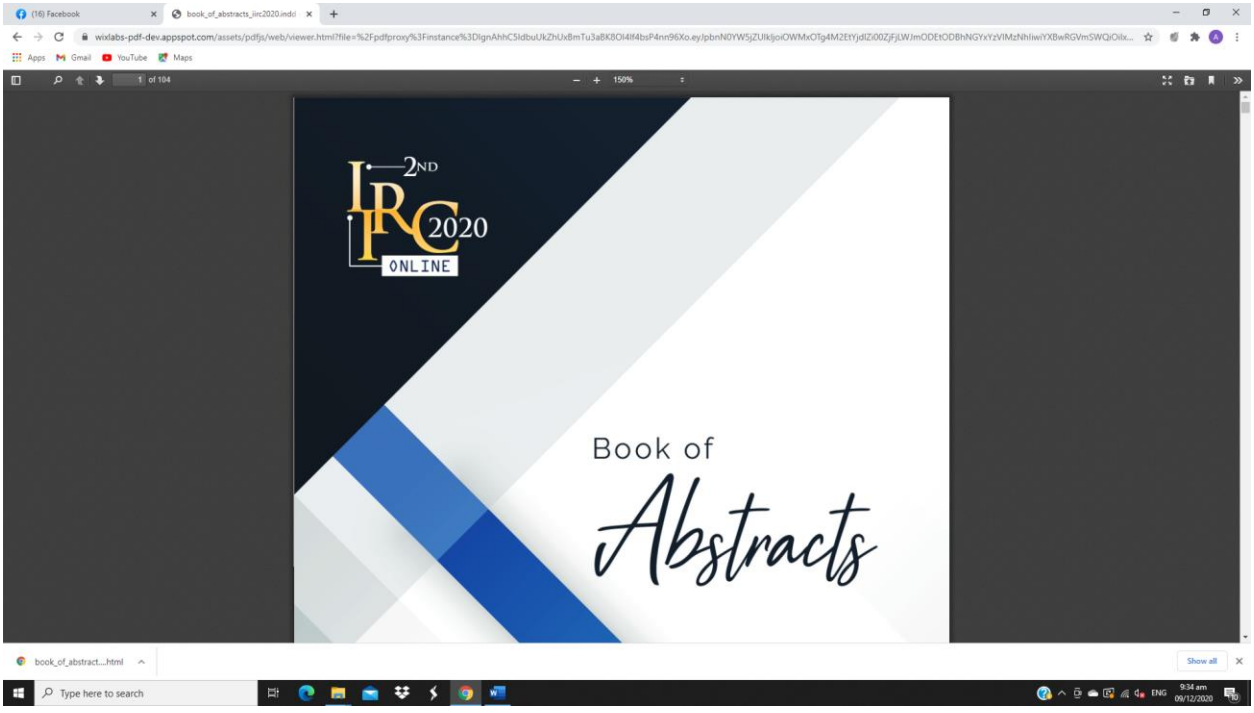

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from the 2020 International
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Research Presentations from the
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# ABIOSORT: THE REPRESENTATIVE-BASED SELECTION SORT ALGORITHM

**Arjay F. Abio**

*Faculty of Partido State University - San Jose Campus*

*Del Carmen, San Jose, Camarines Sur, Philippines*

*abio.arjay@gmail.com*

09099550267

## ABSTRACT

"In computing, sorting is already solved." Probably, many will agree with this statement. However, as sorting is applied to the algorithms, different characteristics arise. These lead to the distinctions of the pros and cons. Thus, various sorting algorithms had been developed, and some are continuously upgrading. The Merge Sort and Quicksort, for instance, are just two of the famous efficient sorting algorithms even in large sizes of data. Yet, due to their recursive structures, they require more memory space compared to those non-recursive ones such as Bubble Sort, Insertion Sort, and Selection Sort. These latter algorithms, on the other hand, are poor in performance in large sizes of data. Hence, when it comes to these algorithms, one significant contribution to computing is to improve their running time performance, which is the objective of this study focusing on the Selection Sort algorithm. This paper presents a new sorting algorithm called AbioSort. It was designed as an enhancement of the original Selection Sort method. While maintaining a being non-recursive structure, it employs a new approach called the "representative-based selection sort" method. Trials revealed that the AbioSort provided an almost 99% improvement in running time performance using posterior analysis. It also demonstrated its 100% accuracy during testing using different sets of data. These show that the AbioSort largely improved the efficiency of the original Selection Sort Algorithm while ensuring accuracy.

*Keywords: algorithm, sorting, representative-based, selection sort*